DATA SCIENCE 1 · 2026–2027

From messy exports to a coach dashboard

Syllabus for labs 3, 4 and 5

Lecturer	dr. Maarten Slembrouck
Session	labs 3, 4 and 5

1 Overview

In labs 3, 4 and 5 you are the new data analyst of a (fictional) youth football and athletics club. The head coach gives you three export files from the club's systems. They are messy on purpose, the way real exports are: inconsistent labels, sprint times in milliseconds, missing values, an athlete who appears twice and a column with a comma as the decimal separator.

Each session builds on the previous one:

Lab 3 **From file to tidy table.** Load, explore, filter, add columns and clean until you can trust the data.

Lab 4 **Combine, summarize, reshape.** Summarize groups, combine tables, switch between a wide and a long shape, and work with a heart-rate series per second.

Lab 5 **Tell the story with plotly.** Choose the right chart for a question, make honest charts with `plotly.express` and combine them into a small interactive dashboard.

2 Learning outcomes

After these three sessions you can:

LO1 load a CSV file with `pandas` and describe its structure, data types and missing values;

LO2 select and filter rows and columns with boolean conditions, and sort the result;

LO3 derive new columns (age, speed, BMI) and convert text to numbers and dates;

LO4 clean a dataset and justify each choice: inconsistent text labels, wrong units, missing values and duplicates;

LO5 summarize data per group with `groupby` and `agg`, and combine tables with `merge` without double counting;

LO6 reshape data between a long and a wide shape with `melt` and `pivot_table`;

LO7 compute basic features of a time series: heart-rate zones, rolling averages and resampling;

LO8 choose a chart type that fits the question, and recognise misleading charts, such as a truncated axis;

LO9 make histograms, box plots, scatter plots and line charts with `plotly.express`, with colour, facets and size for extra dimensions;

LO10 make charts interactive with marimo UI elements and combine them into a dashboard.

3 Prerequisites

Labs 1 and 2 (*Introduction to Python I and II*): variables and data types, operators, `if`/`elif`/`else`, functions, string methods, lists, dictionaries, loops, and reading a CSV file with the `csv` module. Lab 2 ends by summarizing `athletes.csv` with plain Python; in lab 3 you see how `pandas` does the same in a few lines.

4 Software and materials

Every lab is a **marimo notebook** in the `labs/` folder of the course repository. You can open it in two ways:

- **Online:** go to `https://marimo.app`, choose *New* → *Open from URL...* and paste the link to the notebook you get from your lecturer. The data is then read online, so you don't have to install anything.
- **Locally:** install `uv` and Python 3.12 following the *Installing Python* manual (or the slides). Then go to the course folder in a terminal and open a lab with:
  `uv run marimo edit labs/lab3_data_processing_I.py`
  The first time, uv installs the packages it needs; the data is read from the `data/` folder.

Three things to know about marimo:

- Notebooks are **reactive**: when you change a cell, every cell that depends on it runs again, in the order of their dependencies.
- **Two cells cannot define the same variable.** Give every step its own clear name, such as `tests_raw` and `tests_clean`, or keep a sequence of steps together in one cell.
- Every exercise contains a `...` placeholder. A hidden cell below it checks your answer: it turns **green** when the answer is right, and otherwise gives a general hint about what is still wrong, without giving the answer away.

5 The dataset

File	Size	Content and known problems
`athletes.csv`	43 rows, 7 columns	ID, name, birthdate, gender, position, height, weight. 13 spellings for 4 positions, weight as text with a decimal comma (`"57,5"`), one athlete entered twice with two IDs, a few missing values.
`tests.csv`	114 rows, 8 columns	10 m and 30 m sprint, countermovement jump height and Yo-Yo level, at 3 test moments. Some sprint times are stored in milliseconds; a few missing values.
`hr_session.csv`	12,000 rows, 5 columns	Heart rate and speed per second for 5 athletes during one 40-minute training session.

6 Schedule per session

6.1 Lab 3 – Data Processing I: from file to tidy table

Goal: load and explore a dataset, select and filter rows and columns, add new columns and clean the data.

Time	Block	What you practise
13:00–13:20	Warm-up: why not just Excel?	Look at the raw text of a CSV file and see what a table library adds.
13:20–14:00	Loading and exploring	`read_csv`, `.head()`, `.shape`, `.dtypes`, `.info()`, `.describe()`, `mo.ui.table`.
14:00–14:50	Selecting and filtering	Selecting columns, `.loc` with boolean conditions, combining conditions, `sort_values`.
14:50–15:05	Break	
15:05–15:50	New columns	Text to number, age from a birthdate, speed from a sprint time, BMI.
15:50–17:00	Cleaning	Missing values (`dropna` or `fillna`), text labels, units, duplicates, one complete cleaning pipeline.
17:00–17:30	Challenge	The 5 fastest athletes per position. Exit ticket.

Exercises: (1) load `tests.csv`; (2) forwards younger than 16; (3) suspicious sprint times; (4) BMI; (5) age in years; (6) clean the positions; (7) fix the sprint units; (8) remove the duplicate athlete; **challenge:** the 5 fastest per position.

6.2 Lab 4 – Data Processing II: combine, summarize, reshape

Goal: aggregate data with `groupby`, combine tables with `merge`, reshape with `melt` and `pivot`, and compute basic features of a time series.

Time	Block	What you practise
13:00–13:15	Recap: repair a broken cell	Fix a small cleaning pipeline with two bugs.
13:15–14:10	`groupby` + `agg`	Statistics per group, age groups with `pd.cut`, several named aggregations at once.
14:10–14:55	`merge`	Inner and left joins; why a duplicated athlete is counted twice, and why you remove duplicates before merging.
14:55–15:10	Break	
15:10–15:50	`melt` and `pivot`	Long and wide shape; progress between two test moments.
15:50–16:50	Time series	Heart-rate zones as % of HR_{max} (220 − age), time per zone, `rolling`, `resample`.
16:50–17:30	Mini-case in pairs	A training report: one row per athlete.

Exercises: (0) recap: repair the broken cell; (1) group by gender and age group; (2) what did removing the duplicates cost?; (3) progress in jump height; (4) minutes in zone 4–5 per athlete; **mini-case:** training report with the minutes in zone 4–5, the maximum speed and the distance covered per athlete.

6.3 Lab 5 – Data Visualization I: tell the story with plotly

Goal: choose a chart type for a given question, make the standard `plotly.express` charts, add extra dimensions with colour, facets and size, and build in simple interactivity.

Time	Block	What you practise
13:00–13:30	Principles	The right chart for the question (distribution, comparison, relationship, evolution); critique a misleading chart.
13:30–14:30	`plotly.express` basics	Histogram, box plot, scatter plot and line chart, with titles, axis labels and hover information.
14:30–15:15	More dimensions	`color`, `facet_col` and `size`; one fixed colour per position.
15:15–15:30	Break	
15:30–16:15	Heart rate as a time series	Zone bands with `add_hrect`, and why we don't use a second y-axis.
16:15–16:45	Interactivity	`mo.ui.dropdown`, `mo.ui.range_slider`, optionally selections with `mo.ui.plotly`.
16:45–17:30	Mini-dashboard	3 charts and 1 selector that at least one chart responds to.

Exercises: (1) a histogram of weight; (2) Yo-Yo level per position, with facets per gender; (3) speed over time for one athlete; **assignment:** the mini-dashboard.

7 Working method

- Each block starts with a short demo in the notebook, followed by exercises you solve in the notebook itself.
- Use the automatic checks as feedback, not as the goal: when a check turns green, make sure you can explain *why* your code works.
- You do the lab 4 mini-case in pairs.
- Lab 3 ends with an exit ticket: one technique you will reuse and one thing that is still unclear. Bring the unclear points to the next session.

8 Assessment

- **Mini-dashboard (lab 5):** three charts and one interactive selector, made with the club's data. It can serve as your submission for this module and prepares the case presentations and the Project Lab next semester.

9 Preparation per session

- Check before every session that you can open the notebook, locally or on marimo.app, and read the course text for the lab next to the notebook.
- Before lab 3: review lists, dictionaries and functions from lab 2.
- Before lab 4: make sure you can explain every step of the cleaning pipeline at the end of lab 3, because lab 4 starts from it.
- Before lab 5: review `groupby`, `merge` and the long shape from lab 4. Charts with plotly express start from data in the long shape.